

PARKER HONG

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EDUCATION

Northwestern University | Evanston, IL

Anticipated Graduation Jun 2027

McCormick School of Engineering and Applied Science

BS, Manufacturing and Design Engineering

GPA 3.80/4.0

Achievements: High Honors Dean's List (Spring 2025), Dean's List (Fall 2024, Winter 2024, Spring 2024, Winter 2025)

Relevant Coursework: Computer-Integrated Manufacturing, Mechanical Design and Manufacturing, Quality Improvement by Experimental Design, Materials Selection, Mechanics of Materials, Electronics Design, Design Thinking and Communication, Smart Electronics in Design

LEADERSHIP EXPERIENCE

Northwestern Crew | Evanston, IL

Varsity Men's Captain

May 2025 – Present

- Lead a 50+ member intercollegiate club rowing program, coordinating year-long training, competitions, and team logistics.
- Serve as primary liaison between athletes, coaches, and administration, managing team and individual needs and concerns.

Recruitment Chair

May 2024 – May 2025

- Increased athlete participation by 50% through targeted outreach and strategic onboarding initiatives.
- Facilitated overall team cohesion through team-building events and intersquad integration.

MANUFACTURING & ENGINEERING EXPERIENCE

Northwestern Formula Racing – Chassis Engineer

Sep 2025 – Present

- Design a carbon fiber composite steering wheel focused on enhancing ergonomics and driver experience.
- Use rapid prototyping for designs as they evolve, validating safety and compatibility with the driver and vehicle.
- Collaborate with other sub-teams to create a fully functional electric vehicle optimized for performance.

Segal Design Institute – Engineering Intern

Jun 2025 – Dec 2025

- Developed CAD models in SolidWorks and fabricated physical prototypes using CNC, 3D printing, and machining equipment.
- Conducted biweekly design reviews with faculty and industry professionals, incorporating feedback into iterative design improvements.
- Strengthened rapid prototyping skills by transitioning digital concepts to functional prototypes, improving design-for-manufacturability.

Mechanical Design and Manufacturing

Mar 2025 – Jun 2025

- Designed and prototyped a brake caliper system in Siemens NX.
- Performed finite element analysis (FEA) and topology optimization, reducing weight by 44% while maintaining structural integrity.
- Manufactured the final parts with laser-sintered Nylon-12 and validated on a bicycle for a real use-case.

Design Thinking and Communication

Sep 2023 – Jun 2024

- Designed and constructed an elevated mobile nightstand, optimized for dorm room constraints, integrating electrical functionality.
- Created an assistive “money board” for adults with intellectual and developmental disabilities to facilitate independence in use of currency.

EMPLOYMENT EXPERIENCE

Northwestern University Residential Services | Evanston, IL

Sep 2024 – Present

Resident Assistant

- Facilitate acclimation of new students through proactive event planning and intentional engagement of residents.
- Promote a safe, inclusive living environment and act as on-call emergency responder for over 400 residents.
- Provide support and resources while enhancing the overall residential community experience.

Evanston Township High School | Evanston, IL

Jun 2025 – Present

Assistant Varsity Rowing Coach

- Support athletes in developing their rowing technique, fitness, and strategy during water and land workouts.
- Drive team culture and performance by encouraging athletes to maintain focus on team and personal goals.

TECHNICAL SKILLS

CAD and Simulation: SolidWorks, Siemens NX, Onshape, ANSYS, AutoCAD, Rhino, Revit, WUFI

Programming and Analysis: MATLAB, Minitab, Python, HTML/CSS, Java, C, C++, Microsoft Office

Manufacturing and Prototyping: CNC Router, Waterjetting, 3D Printing, Laser Cutting, Milling, Turning, Drilling, Soldering

PROFESSIONAL SKILLS

Personal: Integrity, Discipline, Critical Thinking, Problem-Solving, Goal Orientation, Organization

Interpersonal: Leadership, Collaboration, Communication, Adaptability, Conflict Resolution